



Sardar Patel Institute of Technology
Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

Max. Marks: 60

Class: S.E. (IV) Computer/IT

Name of the Course: Operating systems

ESE Examination

May 2022

Duration: 2hrs

Course Code: CS206/IT206

Instructions:

- (1) All questions are compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Q. No.		Max Marks	CO-BL-PI																					
Q.1 A	State and explain any five operating system services.	5	1-1-1.1.1																					
Q.1 B	Write a short note on multiprogramming system and Time-sharing system with real world example.	5	1-2-1.1.2																					
Q.1 C	Compare between interpreter and compiler.	5	1-2-1.1.2																					
Q.2 A	Define multi-threaded programming with its benefits? Explain the types of threads.	8	2-2-1.1.2																					
Q.2 B	Consider the set of 6 processes whose arrival time and burst time are given below:<table><tr><th>Process ID</th><th>Arrival Time</th><th>Burst Time</th></tr><tr><td>P1</td><td>5</td><td>5</td></tr><tr><td>P2</td><td>4</td><td>6</td></tr><tr><td>P3</td><td>3</td><td>7</td></tr><tr><td>P4</td><td>1</td><td>9</td></tr><tr><td>P5</td><td>2</td><td>2</td></tr><tr><td>P6</td><td>6</td><td>3</td></tr></table> If the CPU scheduling policy is Round Robin with time quantum = 3, Draw the Gantt chart, calculate the average waiting time and average turn-around time. OR	Process ID	Arrival Time	Burst Time	P1	5	5	P2	4	6	P3	3	7	P4	1	9	P5	2	2	P6	6	3	7	3-3-2.4.1
Process ID	Arrival Time	Burst Time																						
P1	5	5																						
P2	4	6																						
P3	3	7																						
P4	1	9																						
P5	2	2																						
P6	6	3																						

	Consider the set of 5 processes whose arrival time, Priority, and burst time is given below: <table><tr><th>Process</th><th>Burst time</th><th>Priority</th><th>Arrival time</th></tr><tr><td>P1</td><td>8</td><td>4</td><td>0</td></tr><tr><td>P2</td><td>6</td><td>1</td><td>2</td></tr><tr><td>P3</td><td>1</td><td>2</td><td>2</td></tr><tr><td>P4</td><td>9</td><td>2</td><td>1</td></tr><tr><td>P5</td><td>3</td><td>3</td><td>3</td></tr></table> Calculate the Average waiting time and Average Turnaround time by using FCFS, Non-preemptive priority, and Priority scheduling algorithms.	Process	Burst time	Priority	Arrival time	P1	8	4	0	P2	6	1	2	P3	1	2	2	P4	9	2	1	P5	3	3	3																																															
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P5	3	3	3																																																																					
Q.3 A	Consider a system that contains five processes P1, P2, P3, P4, P5 and the three resource types A, B and C. Following are the resource types: A has 10, B has 5 and the resource type C has 7 instances. <table><tr><th rowspan="2">Process</th><th colspan="3">Allocation</th><th colspan="3">Max</th><th colspan="3">Available</th></tr><tr><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th></tr><tr><td>P1</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td>3</td><td>3</td><td>2</td></tr><tr><td>P2</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P3</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P4</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P5</td><td>0</td><td>0</td><td>2</td><td>4</td><td>3</td><td>3</td><td></td><td></td><td></td></tr></table> 1. What is the content of the need matrix? 2. Determine if the system is safe or not. If yes, give the order in which process should run to completion. 3. What will happen if the resource request (1, 0, 0) for process P1 can the system accept this request immediately. justify?	Process	Allocation			Max			Available			A	B	C	A	B	C	A	B	C	P1	0	1	0	7	5	3	3	3	2	P2	2	0	0	3	2	2				P3	3	0	2	9	0	2				P4	2	1	1	2	2	2				P5	0	0	2	4	3	3				8	3-3-2.4.1
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P5	0	0	2	4	3	3																																																																		
Q.3 B	Define critical section problem. Explain Peterson's solution with pseudocode to solve critical section problem.	7	3-3-2.4.1																																																																					
Q.4 A	Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The head is initially at cylinder number 53 moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. What is the total head movement (in number of cylinders) incurred while servicing these requests for FCFS (First come first serve) and SSTF (shortest seek time first) algorithms? OR Define Inode structure. A file system uses UNIX inode data structure which contain 8 direct block addresses, one indirect block, one double and one triple indirect block. The size of each disk block is 128bytes and size of each block address is 8byte. Find the max. possible file size?	5	5-3-2.4.1																																																																					
Q.4 B	What is dining philosopher problem, explain its solution with pseudocode using semaphore.	5	3-3-3.2.2																																																																					
Q.4 C	What is variable size partitioning? Explain its advantages and disadvantages. (Any two) OR Explain dynamic memory allocation with example? Given 5 partition of 100kb, 500kb, 200kb, 300kb, and 600kb (in order), use the first fit, best fit, and worst fit algorithms to place 212kb, 417kb, 112kb, and 426kb (in order)	5	4-2-1.1.2 4-3-2.4.1																																																																					